Comparison of the modified biophysical profile to a "new" biophysical profile incorporating the middle cerebral artery to umbilical artery velocity flow systolic/diastolic ratio

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OBJECTIVES: The objective of this study was to determine whether the addition of the middle cerebral to umbilical artery systolic/diastolic velocity waveform ratio to the modified biophysical profile would improve perinatal outcome in patients at high risk.

STUDY DESIGN: A prospective, randomized outcome study of patients referred to the perinatal laboratory for antenatal surveillance was undertaken. Six hundred sixty-five patients were randomized to two antenatal surveillance protocols: group 1, modified biophysical profile; and group 2, modified biophysical profile plus evaluation of the middle cerebral artery to umbilical artery systolic/diastolic ratio. Patients were followed up serially and neonatal outcome data including gestational age at delivery, birth weight, incidence of cesarean section delivery for fetal distress, admission to the neonatal intensive care unit, days in the neonatal intensive care unit, and the presence of significant neonatal morbidity were tabulated.

RESULTS: The total population showed no statistical difference in outcome parameters between groups 1 and 2. However, a subgroup of patients evaluated for suspected uteroplacental insufficiency did show a significant reduction in caesarean section for fetal distress in group 2 patients.

CONCLUSIONS: In a subgroup of patients at risk for uteroplacental insufficiency, the addition of the middle cerebral/umbilical artery ratio to an antenatal surveillance protocol should be expected to improve perinatal outcome. (Am J Obstet Gynecol 1998;178:1346-53.)

Key words: Middle cerebral artery, umbilical artery, antenatal surveillance, modified biophysical profile, Doppler ultrasonography

Many investigators believe that the use of antenatal fetal surveillance tests has led to a significant decrease in perinatal mortality and morbidity rates.[1-3] The most common test used for antenatal fetal evaluation is the nonstress test (NST), either alone or in combination with other fetal parameters in the form of a fetal biophysical profile.[1-4] The modified biophysical profile (NST and fluid volume only), the primary surveillance test used at our institution, appears to be as accurate a predictor of perinatal outcome as the full biophysical profile.[5]

The NST is considered the most sensitive indicator of fetal central nervous system oxygenation, and, according to the hypothesis of Vintzileos et al.,[4] is the first parameter of the biophysical profile to become abnormal with developing fetal hypoxia. Some recent studies suggest that a nonreactive NST, however, may be a very late sign of fetal compromise.[6-9]

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Work at our own institution and reports of other investigators suggest that additional fetal evaluation with pulsed Doppler velocity blood flow wave analysis may provide further situations.[10-15] Comparison of information concerning fetal status in high-risk systolic/diastolic ratios (S/D ratio) in the middle cerebral artery to the S/D ratio in the umbilical artery appears to be an accurate method for prediction of fetal outcome, especially in the growth-restricted fetus.[10, 11, 16-19]

A pilot study at our institution of 171 patients at high risk who had both NST and middle cerebral artery/umbilical artery ratio within 1 week of delivery suggested that the ratio was a more sensitive indicator of fetal compromise than the NST.[20] A prospective, randomized, clinical outcome study was undertaken to test the hypothesis that the addition of the ratio to the modified biophysical profile would improve the ability to identify fetuses at risk for poor perinatal outcome and thereby decrease neonatal morbidity rates.

Material and methods

All patients referred to the perinatal laboratory at our institution between June 1, 1995, and Nov. 1, 1996, for a modified biophysical profile were offered an opportunity to participate in the study. Patients with multiple gesta-

Table I. Significant neonatal complications (excluding respiratory distress syndrome)

Complication	*Group 1*	*Group 2*	*Total*
CNS complications	4	2	6
Sepsis	9	12	21
Acidosis/asphyxia	3	8	11
Cardiomyopathy	2	2	4
Anemia	2	0	2
Metabolic	2	3	5
TOTAL			49 (7.4%)

CNS, Central nervous system.

Table II. Indications for testing

Indication	*Group 1*	*Group 2*
Risk of UPI*	56	63
Fetal risk†	92	97
Postdates	71	70
Maternal diabetes	26	24
PROM/PTL	34	39
Fluid abnormalities	15	16
Other	23	39

UPI, Uteroplacental insufficiency; *PROM,* premature rupture of membranes; *PTL,* preterm labor. $\chi^2_{df6} = 3.70$, $p = 0.7175$, not significant.

*Risk of uteroplacental insufficiency: Hypertensive disease, collagen-vascular disease, significant drug abuse.

†Fetal risk: Decreased fetal movement, poor obstetric history, suspicious NST in outside office, fetal arrhythmia, suspected small for gestational age with normal umbilical artery Doppler studies.

tion or fetuses with known chromosomal or significant structural abnormalities were excluded from the study. Patients who agreed to participate were randomized into one of two groups on the basis of a computer-generated, random-number allocation system.

Group 1 patients underwent weekly modified biophysical testing with NST and amniotic fluid determination with the use of the four-quadrant, vertical pocket technique. A reactive NST was defined as the presence of two or more fetal heart rate accelerations of 15 beats or more, within a 10-minute window, lasting at least 15 seconds and associated with fetal movement.[1] Patients with nonreactive NSTs were evaluated by one of the maternal fetal medicine specialists and either retested within 24 hours or were recommended for delivery. If amniotic fluid volume index was <5 cm, the patient was retested within 2 to 3 days. In patients with ultrasonic evidence of growth restriction (estimated weight below the 10th percentile) Doppler velocity blood flow analysis of the umbilical artery was also recommended. Both the physicians involved in the study and the institutional review board of the hospital believed that there was sufficient evidence that Doppler velocity blood flow studies of the umbilical artery were of clinical benefit to patients at high risk and that these studies should be available to patients in the control group (group 1) if their referring physician requested it.

Group 2 patients also underwent modified biophysical testing as described above but also had the middle cerebral artery/umbilical artery ratio calculated. The umbilical artery S/D ratio was calculated from three or more successive waveforms obtained from a free-floating portion of the umbilical cord during minimal fetal activity and the absence of fetal breathing. The middle cerebral artery S/D ratio was measured in a similar fashion just distal to the junction of the middle cerebral and internal carotid arteries. The ratio between the middle cerebral S/D ratio and umbilical artery S/D ratio was calculated, and on the basis of previous published studies, a value <1.0 was considered abnormal.[10, 11] If there was absent or reverse flow in the umbilical artery the middle cerebral artery/umbilical artery ratio was assigned a value of 0. If the NST, fluid volume index, and middle cerebral artery/umbilical were all normal, the patient was retested in 1 week. If any one of these parameters were abnormal, the patient was retested in 2 to 3 days. If any two of these parameters were abnormal the patient was evaluated by one of the maternal fetal medicine specialists and either retested within 24 hours or advised of the need for delivery. Patients with all three tests abnormal were admitted to the hospital for evaluation for delivery.

Clinical outcome parameters were evaluated for each group after delivery. Neonatal morbidity rate was evaluated by three parameters: the need for admission to the neonatal intensive care unit; length of stay in days in the neonatal intensive care unit; and the presence or absence of significant neonatal morbidity. Significant neonatal morbidity was defined as the presence of central nervous system complications such as significant intraventricular hemorrhage, anoxic encephalopathy, severe retinopathy, or seizures; the presence of necrotizing enterocolitis; the presence of early neonatal sepsis; or other significant complications as listed in Table I. Respiratory distress was not included because this complication was believed to be primarily related to gestational age. Secondary outcome parameters including gestational age at delivery, neonatal weight, and delivery by cesarean section for fetal distress in labor were also tabulated. For the purposes of this study, fetal distress in labor was defined as the presence of repetitive late decelerations, persistent loss of baseline variability, or repetitive, severe variable decelerations.

Data were analyzed for both the entire patient population and for a subgroup of patients who were believed to be at risk for uteroplacental insufficiency caused by hypertensive disease, collagen-vascular disease, or significant drug abuse during pregnancy (see Table II). Statistical comparisons were made for neonatal outcome between groups by use of χ^2 analysis or Fisher's exact test for discrete data or a two-sample Student t test for continuous data. A value of $p < 0.05$ was considered to be sig-

Table III. Demographic and general outcome data

	Group 1	Group 2	Statistical significance
Maternal age (yr)	28.7 (6.6)	29.1 (6.2)	$p = 0.3739$
Mean fluid index at last test (cm)	15.0 (5.8)	14.7 (5.8)	$p = 0.5095$
Duration test to delivery (days)	14.0 (17.3)	9.3 (15.8)	$p = 0.0003$
Nonreactive NST	3.2%	1.7%	$p = 0.2292$
Gestational age at delivery (wk)	38.8 (2.5)	39.0 (3.5)	$p = 0.4574$
Birth weight (gm)	3337 (680)	3289 (348)	$p = 0.3629$
Cesarean section for fetal distress	3.8%	4.0%	$p = 0.8746$

Table IV. Neonatal morbidity rates: All patients

	Group 1	Group 2	Statistical significance
Admission to neonatal intensive care unit	49	60	$p = 0.5348$
Length of stay in neonatal intensive care unit (days)	1.7 (6.6)	1.7 (7.4)	$p = 0.9666$
Significant neonatal complications	6.6%	8.0%	$p = 0.5891$

nificant for all the above tests. Before the study, a power analysis with an α error of 0.05 and a β error of 0.20 (80% power) was undertaken to determine the number of patients needed in each arm of the study.[21, 22] On the basis of the incidence of significant neonatal complications of 15% seen in our pilot study, the power analysis indicated that 282 patients would be needed in each group to show a significant reduction in neonatal complications between the two groups. This study was approved by the hospital institutional review board.

Results

During the study period, there were 899 patients with singleton pregnancies referred to the perinatal laboratory for modified biophysical testing that met the inclusion criterion. One hundred eighty-four (20.5%) refused participation in the study, leaving 715 patients randomized for participation in the study. Of these patients, 50 (7.0%) withdrew from the study, delivered at another institution or were lost to follow-up, leaving 665 patients for analysis. Table II lists the indications for testing. Analysis by χ^2 showed no significant differences between the two groups in respect to indications for testing ($\chi^2 = 3.7_{df6}$, $p = 0.7175$, not significant).

General demographic and outcome data for each group are shown in Table III. There were no significant differences in maternal age, mean amniotic fluid index, percentage of patients with nonreactive NST, gestational age at delivery, birth weight, or the percentage of patients delivered by cesarean section for fetal distress between the two groups. There was, however, a statistically longer interval between the last test and delivery in group 1 compared with that in group 2. Despite the protocol recommending weekly intervals between testing, many of the attending physicians did not follow this recommendation and only requested retesting at weekly intervals when the patient was close to term.

Table IV shows the neonatal morbidity rates for each group and indicates that there were no significant differences in neonatal morbidity rates between the two groups. There were two perinatal deaths, one in each group: in group 1 there was a 1760 gm infant delivered at 33 weeks of gestation after a nonreactive NST and oligohydramnios the day of delivery. The patient had been referred for evaluation of fetal arrhythmia, and the diagnosis of Epstein anomaly had been made at the time of examination. The neonate died of cardiac failure and suspected sepsis. In group 2 there was a 2500 gm infant delivered at 36 weeks of gestation after preterm labor who died from a rapid-onset group B streptococcal infection. This patient had a reactive NST and normal middle cerebral artery/umbilical artery ratio 5 days before delivery.

Table V shows the outcome statistics for the subgroup of patients believed to be at risk for uteroplacental insufficiency. As with the entire group of patients, there were no significant differences in maternal age, mean amniotic fluid index, percent of patients with nonreactive NST, gestational age at delivery, or birth weight between the two groups. There was also a statistically longer interval between the last test and delivery in group 1 compared with group 2, and there was a statistically higher incidence of cesarean delivery for fetal distress in group 1 compared with group 2.

Comment

The results of this prospective, randomized outcome study seem to indicate that the addition of the middle cerebral artery/umbilical artery ratio to the modified biophysical profile does not improve neonatal outcome in the general high-risk population. However, in a subgroup of patients at risk for uteroplacental insufficiency, the addition of the middle cerebral artery/umbilical artery ratio to the antenatal testing protocol was associ-

Table V. Neonatal morbidity: Patients at risk for uteroplacental insufficiency

	Group 1	*Group 2*	*Statistical significance*
No.	56	63	
Maternal age (yr)	30.0 (7.1)	31.0 (5.7)	$p = 0.4620$
Mean fluid index at last test (cm)	15.3 (5.9)	14.5 (5.2)	$p = 0.4969$
Duration test to delivery (days)	14.2 (18.1)	6.2 (13.4)	$p = 0.0072$
Nonreactive NST	8.9%	3.2%	$p = 0.1738$
Gestational age at delivery (wk)	37.6 (2.7)	37.7 (2.3)	$p = 0.8388$
Birth weight (gm)	3057 (741)	3086 (771)	$p = 0.8129$
Cesarean section for fetal distress	6 (10.7%)	1 (1.6%)	$p = 0.0406$
Admission to neonatal intensive care unit	14	12	$p = 0.4328$
Length of stay in neonatal intensive care unit (days)	3.8 (10.4)	2.4 (7.0)	$p = 0.3973$
Significant neonatal complications	8.6%	7.9%	$p = 0.9342$

ated with a decrease in the incidence of cesarean section for fetal distress; although not statistically significant, this was also associated with a 1-day decreased stay for infants admitted to the neonatal intensive care unit. This suggests that in patients at risk for uteroplacental insufficiency the middle cerebral artery/umbilical artery ratio will identify fetal compromise earlier than the NST. This conclusion is supported by numerous studies that have shown the value of Doppler velocity flow studies in the diagnosis and management of intrauterine growth retardation.[10, 12, 16, 23, 24]

Other investigators have compared the ability of the NST and middle cerebral artery Doppler velocity studies to evaluate fetal compromise. Weiner et al.[25] serially evaluated 13 fetuses with absent end-diastolic velocity in the umbilical artery until delivery and found that fetal heart rate patterns remained normal despite significant increases in diastolic velocity flow in the middle cerebral artery. Fetal heart rate abnormalities did not occur until the middle cerebral artery lost its ability to vasodilate. Chandran et al.[26] serially followed 27 patients at high risk with middle cerebral artery Doppler velocity studies and NST. They found the hypoxemia (as defined by abnormal umbilical artery blood gas studies at birth) was better recognized by the fetal middle cerebral artery flow velocity than by fetal heart rate analysis. Guzman et al.[27] evaluated 32 growth-restricted fetuses with both middle cerebral artery Doppler velocity flow studies and NST and found that the former was a better predictor of acid-base status at birth. Devine et al.[18] compared the ability of the middle cerebral artery/umbilical artery ratio, NST, or biophysical profile to predict adverse perinatal outcome in 49 postdate pregnancies and found that the middle cerebral artery/umbilical artery ratio was a better predictor of adverse outcome than the other two tests. Other studies have compared other Doppler velocity indices with the NST or biophysical profile and also found that Doppler studies are slightly more outcome sensitive that than those tests for predicting poor perinatal results.[28, 29]

The current study is a large, prospective, randomized study of the use the middle cerebral artery/umbilical artery ratio, together with the NST, for fetal surveillance. Overall, there did not appear to be any significant improvement in neonatal outcome in the middle cerebral artery/umbilical artery ratio group. However, in the subgroup patients evaluated for suspected uteroplacental insufficiency, the addition of the ratio to the antenatal surveillance protocol was associated with fewer caesarean sections for fetal distress and, though not statistically significant, an average of 1 day less stay in the neonatal intensive care unit. Because of the small number of patients in this subgroup, the statistical analysis of neonatal outcome lacked sufficient power to prove that this 1-day difference was significant. If larger studies confirm that this 1-day decrease in length of stay is significant, then the cost savings and potential decreased severity of neonatal compromise would indicate that the addition of the middle cerebral ratio/umbilical artery ratio to antenatal surveillance would be beneficial.

Finally, there are some hidden biases in the study that interfere with the interpretation of the results. Although the a priori power analysis indicated that there were adequate numbers of patients in each arm of the study, this analysis was based on an assumed incidence of significant neonatal complications of 15%, derived from previous studies.[20] In this study, however, the overall significant neonatal complication rate was only 7.4%. Power analysis indicates that 601 patients would be needed in each group to avoid β error. The significant difference in duration from last test to delivery (14 days for group 1 versus 9.3 days for group 2) suggests that patients in group 1 were less likely to follow the suggested protocol of weekly testing, and this may have biased the outcome. Finally, because Doppler velocity blood flow studies were available to patients in the control group (group 1) if their referring physician requested it, potential bias is introduced into the study. Thirty-seven (11.7%) of the total patients in group 1 and 9 (16%) of the subgroup of patients in group 1 with suspected uteroplacental insufficiency had Doppler velocity flow studies in addition to the NST for antenatal surveillance. The problems of attending physician compliance to protocols and its adverse effects on the outcome of a randomized clinical trial of Doppler ultrasonography have been discussed by Nienhuis et al.[30]

In summary, this study shows small but important improvement in neonatal outcome for patients at risk for uteroplacental insufficiency when the middle cerebral artery/umbilical artery ratio is added to a protocol of antenatal surveillance. Larger studies of patients at high risk in this subgroup will be needed to confirm these results.

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Discussion

DR. GEORGE R. SAADE, Galveston, Texas. I congratulate Ott et al. on this large, randomized study evaluating the usefulness of adding Doppler ultrasonographic studies to the modified biophysical profile consisting of an NST plus an amniotic fluid index. Before I comment on the conclusion of the study, I would like to discuss a few issues and pose a few questions specific to the report provided to me before the meeting.

Doppler studies were made available to the patients randomized to the modified biophysical profile–only group whenever the referring physician requested it or if ultrasonographic evidence of fetal growth restriction was present. How many patients in the biophysical profile–only group did receive Doppler studies, and what effect could this have had on the results? Did the study in fact compare routine with selective addition of Doppler ultrasonographic studies with the modified biophysical profile instead of comparing a "new" biophysical profile with the modified biophysical profile?

Neonatal respiratory distress was excluded from the criteria for "significant neonatal morbidity" because it was believed to be primarily related to gestational age. Why then were intraventricular hemorrhage, severe retinopathy, necrotizing enterocolitis, and early neonatal sepsis, all of which are commonly associated with prematurity, included?

In the calculations for the sample size, the investigators found that 282 patients would be needed in each group to show a "significant" reduction in the selected neonatal complications. Evaluation of the mostly negative findings of this study requires knowledge of what the authors considered "significant" reduction. What was the actual number used in the sample size calculation?

The authors concluded that the addition of Doppler ultrasonographic studies to the modified biophysical profile should be expected to improve perinatal outcome in a subgroup of patients at risk for uteroplacental insufficiency. This conclusion was based on the findings that there were no statistically significant differences in the outcome measures when all the patients were considered, but there was a significant decrease in the incidence of cesarean section for fetal distress and a small albeit nonsignificant, decrease in the neonatal intensive care unit length of stay in the subgroup of patients at risk of uteroplacental insufficiency who were randomized to the Doppler ultrasonography group compared with those who were randomized to the modified biophysical profile only. But is a 1-day difference in neonatal intensive care stay clinically important when the SDs were as large as 7 to 10 days? As for the other outcome measure, the authors do not provide us with the incidence of cesarean section for other indications. Was the decrease in cesarean section for fetal distress offset by an increase in other indications? Was the decrease in this outcome measure achieved at the expense of more cesarean sections without trial of labor performed for abnormal Doppler ultrasonographic studies? Because fetal distress is a very subjective measure, were the cases reviewed by a blinded investigator? Were cord blood gases obtained? More important, does a decrease in cesarean sections for fetal distress equal improved perinatal outcome? Furthermore, neither of the other measures of perinatal outcome studied, mainly admission to neonatal intensive care unit and "significant" neonatal complications, was significantly different between the two groups.

The most significant differences, however, were in the number of days between last antenatal test and delivery in the overall, as well as in the subgroup analysis. These differences were much more significant than the differences in cesarean section for fetal distress or length of neonatal intensive care unit stay in either analysis. The group of patients randomized to modified biophysical profile only had a mean of more than 14 days between the last test and delivery compared with a mean closer to 7 days in the study group in both the original and subgroup analyses. Compared with the group of patients randomized to the "new" biophysical profile, a significantly larger number of patients in the control group did not receive weekly testing as originally planned. This may have been caused by differences in patient compliance, physician preferences, or level of care (patients who received Doppler ultrasonography may have been evaluated by subspecialists or more senior providers or their cases were scrutinized more carefully). Was the study then biased in favor of the "new" biophysical profile? An alternative reason for these differences is that the additional test in the study group, in this case the Doppler ultrasonography, may have increased the chance of abnormal findings and led to more frequent testing. Does the difference in this parameter between the two groups outweigh the small difference in a questionable outcome measure? If so, would it be appropriate to use an alternative conclusion that patients receiving Doppler ultrasonographic studies in addition to biophysical profile undergo more frequent testing, without a clear benefit to perinatal outcome?

Dr. Frank C. Miller, Lexington, Kentucky. The goal of antepartum testing is to reduce the incidence of perinatal morbidity and mortality rates. It is generally agreed that the most frequently used screening test is the modified biophysical profile, which consists of a non-stress test and an estimate of amniotic fluid volume, that is, the amniotic fluid index. If the modified biophysical profile is abnormal, back-up tests usually include either a full biophysical profile or a contraction stress test. Dr. Ott et al. have elected to study the value of a different test, that is, the fetal cerebral artery S/D ratio compared with the umbilical artery, S/D ratio in combination with the modified biophysical profile as the antenatal tests for patients at high risk. The rationale for studying the middle cerebral artery/umbilical artery S/D ratio is that in postdate pregnancy and fetuses with growth restriction, Doppler ultrasonography demonstrates a brain-sparing affect in the presence of hypoxia. Investigators have shown a low index of resistance in the middle cerebral artery with fetal compromise. A ratio of middle cerebral artery/umbilical artery Doppler values has been proposed as a better predictor of adverse pregnancy outcomes than the Doppler value of either vessel considered alone. A prospective, randomized trial was conducted on all patients referred to the authors' antenatal testing service. Patients were randomized into two groups. Group 1, the control group, had weekly modified biophysical profiles. Patients with nonreactive NST were retested in 24 hours or were delivered. An amniotic fluid index of <5 resulted in retesting in 2 to 3 days. Although not part of the study protocol, any fetus diagnosed as growth restricted had Doppler velocity blood flow analysis of the umbilical artery. The report does not address how a patient with an abnormal test was managed.

Group 2, the study group, had a modified biophysical profile and the middle cerebral artery/umbilical artery ratio (values less than 1 were considered abnormal). Any one abnormal test triggered retesting in 2 to 3 days, two abnormal tests resulted in retesting in 24 hours, and three abnormal tests were grounds for admission and delivery. It was not clear in the report whether all three tests were assigned the same clinical value; for example, was an abnormal NST clinically more important that a low amniotic fluid index? This is an important distinction because in group 1, abnormal NSTs were retested at shorter intervals than low amniotic fluid index (i.e., 24 hours versus 2 to 3 days).

In the study design, a power analysis was conducted to determine the size of the study population. On the basis of prior experience, the authors estimated that the significant neonatal complication rate would be approximately 15%. Surprisingly, during this study period they found a neonatal complication rate of only 7.4%. This rate would have necessitated a much larger study population in order to show a significant difference between the two groups. In a California study of 2774 patients with more than 17,425 antenatal tests, Nageotte et al.[1] reported a similar neonatal complication rate of 7% in their patient population.

The authors found no differences in neonatal morbidity rates between their two study groups. The small subcategory of patients at risk for uteroplacental insufficiency had too few study patients to make valid conclusions regarding the added value of Doppler flow measurements. Subgroup 1 (uteroplacental insufficiency) had more nonreactive NSTs, and the time from the last test to delivery was about 2 weeks compared with 1 week in group 2. There were more cesarean deliveries for fetal distress in group 1, but there were also more than twice as many nonreactive NSTs in group 1. The longer stay in the neonatal nursery for group 1 neonates could have been due to problems not measured by the antenatal testing. With this small number of patients, one neonate with a long stay not related to perinatal events could account for the difference.

In summary, I congratulate the authors on a good preliminary study. They found no differences in the outcomes between the control and study groups. As they concluded, larger studies of the patients at high risk in the uteroplacental subgroup will be needed to reach any conclusions about the additional value of the middle cerebral artery/umbilical artery ratio in that group.

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Dr. Robert J. Carpenter, Houston, Texas. Unless I missed it somewhere, I never saw absolute numbers of abnormal S/D ratios. I never saw any information about those individuals with reversal or absence of flow. I would like to know what those are and also the gestational ages of those particular fetuses versus the rest because I was markedly unimpressed by any of the statistical data.

Dr. Dwight P. Cruikshank, Milwaukee, Wisconsin. I am curious about the patient population. You talked about a subgroup that you split off who were at risk for uteroplacental insufficiency. My understanding of antepartum testing is, that is who we are supposed to do it on. We are looking for stillbirths—trying to prevent stillbirths in patients at risk for uteroplacental insufficiency. So these 550 patients who were not at risk for that—why were they being tested?

Dr. Yoram Sorokin, Detroit, Michigan. I have a suggestion for the study. On the basis of the literature, the problem, I think, is with the numbers and the selection of patients, because the randomized trials of umbilical artery velocimetry when they looked at the β analysis found that actually it was significant when they got to about 6000 patients that were randomized in several trials and only for patients with fetal growth restriction and preeclampsia.

So there is the potential that the velocimetry will be helpful, but to do that one would have to do a study that would have a very large number of patients and select those that have fetal growth restriction as well as preeclampsia.

The other comment I still have to make is despite all the studies we have in the literature, there is no study of any antenatal testing, any antenatal testing that has been randomized against placebo that has shown any benefits so far.

Dr. James Bofield, Dayton, Ohio. If I am not mistaken, this is the same group who recently published a cervical ripening study that noted that private physicians essentially perform cesarean deliveries very, very frequently for any fetal heart rate abnormality; so I would urge caution in the interpretation of this cesarean section data unless there is a lot more investigation into whether the cesarean sections actually needed to be performed. We need things such as cord arterial pH and so on.

Dr. Ott (Closing). The first question was about how many patients in the control group had Doppler ultrasonography, and as I mentioned 11% did.

This was umbilical artery and not middle cerebral artery. Most of these patients had also fetal weight estimations; and so if they had estimated fetal weight below the 10th percentile, they had umbilical artery Doppler studies also done. During this whole study period, in the control group the criteria for delivery was the nonstress test and our amniotic fluid volume, not the Doppler studies. Respiratory distress syndrome was not included in the severe neonatal complications because it was believed to be, as I mentioned, caused by gestational age, not morbidity per se; indeed, all these other complications that were included also are highly dependent on gestational age, but they also do occur in later gestation. So it was believed to be appropriate to include those in the neonatal complications.

There is really no good definition of what neonatal morbidity is. Most studies use their own, and we fell back on that. A lot of the comments concerning cesarean section—and I would fully and completely agree that the numbers in this subset are too small to make firm conclusions about whether or not this testing procedure will improve perinatal outcome in that respect.

The other statistics, as was mentioned by the discussant and the comments from the floor, did not show any improvement in neonatal outcome in any of the other parameters, but we believe that this study does point to the need for a concentration on those patients specifically at risk for uteroplacental insufficiency (that is, hypertensive patients or growth-restricted patients who have evidence of growth retardation or other complications), and we will plan to continue our studies in that area and try to recruit other collaborative people to do these studies.

Why was there a significantly longer test interval in the study group as compared with the control group? That, of course, is one of the big problems in doing large clinical studies in a private patient population. As I mentioned, we just do not have good enough control over these patients.

A number of these patients were sent in, I freely admit, for soft causes. A reactive nonstress test was determined, and the private physician decided not to send the patient back despite our continued encouragement for them to maintain the protocol. So as I mentioned, this does interject bias into the study, and it is one of the difficulties in doing large clinical studies.

I am sure that I missed a lot of other comments, but I would like to sum up.

Unidentified speaker. The absolute numbers of the abnormals?

Dr. Ott. I don't have that right at my fingertips. I do have the data in our database, but again, we were not using the umbilical artery systolic/diastolic ratio or other ratios per se but were using the ratio between the middle cerebral and the umbilical artery as our criteria. If the umbilical artery ratio goes to zero or reverses, then the ratio is considered to be totally abnormal and that would be in the abnormal group.